

Bhavya Patel | AI/ML Engineer

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SUMMARY

AI/ML Engineer with 3 years of experience building Generative AI, Agentic workflows, and computer vision solutions across FinTech, enterprise content, and research environments. Proven track record in developing RAG pipelines, LLM orchestration, semantic caching, and automated evaluation using Python, LangGraph, AWS Bedrock, Azure ML, and MLflow. Skilled in productionizing deep learning models with vector databases, REST APIs, and Docker-driven CI/CD automation.

SKILLS

Generative AI & LLMs: Agentic Workflows, LangGraph, LangChain, RAG, Multi-Stage RAG, Prompt Engineering, Prompt Evaluation, Semantic Search, Embeddings, Semantic Caching, Hugging Face Transformers

Machine Learning & Deep Learning: Machine Learning, Deep Learning, NLP, Computer Vision, CNNs, Vision Transformers (ViT), Scikit-learn, TensorFlow, TensorFlow Hub, PyTorch, Model Training, Model Evaluation, Hyperparameter Tuning, Feature Engineering, Image Processing

Data & Vector Systems: Pinecone, ChromaDB, Vector Databases, Vector Search, Data Pipelines, Pandas, NumPy, SQL

Cloud & MLOps: AWS Bedrock, Azure Machine Learning, MLflow, MLOps, Model Monitoring, Experiment Tracking, Model Deployment, Docker, Kubernetes, CI/CD, GitHub Actions

Programming & Software Engineering: Python, Java, Shell Scripting, Linux, REST APIs, FastAPI, Distributed Computing, API Integration, Git, GitHub, Streamlit

PROFESSIONAL EXPERIENCE

AI/ML Engineer | State Street, Canada (Remote) Jun 2026 – Present

- Architected multi-stage RAG workflows with LangGraph and LangChain across secure financial repositories, reducing compliance-document retrieval latency by 31% through optimized retrieval sequencing and context selection.
- Implemented semantic caching and dynamic token routing across AWS Bedrock and Azure ML inference workflows, reducing redundant model processing by 27% while preserving required compliance context.
- Automated enterprise compliance-document auditing through Agentic AI workflows, converting manual review bottlenecks into auditable processing pipelines and increasing document-processing throughput by 34%.
- Optimized LLM prompt evaluation and inference workflows to control cloud operational overhead, reducing average token consumption per audited document by 22% across production evaluation runs.
- Integrated MLflow monitoring with GitHub Actions CI/CD to deploy Docker-containerized AI services across scalable cloud infrastructure, establishing repeatable evaluation and deployment gates across a 6-person Toronto-Boston engineering team.

Teaching Assistant – Distributed Systems | Concordia University, Canada Sep 2025 – May 2026

- Engineered Java RMI and CORBA lab implementations for distributed communication exercises, reducing repeated student runtime failures by 28% through reusable service and debugging templates.
- Optimized SOAP-based Web Service examples and network communication workflows, improving average laboratory execution time by 24% through streamlined service invocation and error-handling patterns.
- Automated grading workflows with shell scripting to reduce repetitive evaluation effort by 32%, enabling faster feedback cycles across weekly distributed-systems assignments.
- Diagnosed network communication bottlenecks across student implementations, reducing recurring troubleshooting cases by 26% through structured debugging procedures and reusable technical documentation.
- Designed hands-on distributed computing laboratories for a cohort of 20+ graduate students, translating multi-node architecture concepts into executable Java-based exercises and technical troubleshooting sessions.

Data Science Intern | Blue Data Consulting & IT Services, India Dec 2023 – Jun 2024

- Engineered a generative content pipeline using Python, FastAPI, Streamlit, Pinecone, ChromaDB, and LLM workflows, increasing automated content-processing efficiency by 40% across enterprise learning assets.
- Optimized vectorization and retrieval workflows for legacy enterprise content, reducing average content retrieval latency by 29% through improved indexing and semantic-search execution.
- Automated structured video-learning content generation from legacy enterprise assets, reducing manual content transformation effort by 35% through GPT-3.5-Turbo prompt and workflow optimization.
- Benchmarked Hugging Face and commercial LLM configurations across latency and output-quality dimensions, improving selected model-response accuracy by 18% through systematic prompt evaluation.

Machine Learning Research Student | CHARUSAT University, India Mar 2022 – Jun 2023

- Engineered a NumPy and Pandas video-processing pipeline that curated more than 13,000 UCF-101 frames, streamlining dataset preparation through automated preprocessing.
- Optimized Vision Transformer training with TensorFlow and TensorFlow Hub, achieving 94% classification accuracy through systematic hyperparameter tuning and image-augmentation experiments.
- Improved multi-sport activity recognition performance by transitioning experimental workflows from CNN-based architectures to Vision Transformers, increasing classification accuracy by 12 percentage points.
- Evaluated Vision Transformer and CNN-based activity-recognition models using TensorFlow and scikit-learn evaluation workflows, achieving 94% classification accuracy through systematic hyperparameter tuning and image-augmentation experiments.
- Collaborated with 2 peer researchers and a Principal Research Investigator to structure video datasets, execute experiments, and validate computer-vision model performance across activity-recognition workflows.
- Prepared experimental findings for publication in the Springer Book Series, documenting model methodology, dataset preparation, evaluation results, and comparative computer-vision performance.

Flutter Development Trainee | Inspire Cyber Security, India May 2022 – Oct 2022

- Engineered responsive Flutter interfaces from Adobe XD wireframes, establishing reusable component structures and consistent application layouts.
- Integrated asynchronous REST APIs and local database operations within Flutter workflows, reducing average screen data-loading time by 21% through optimized state and request handling.
- Developed the Palm Box Cricket Booking System with Flutter and Dart, implementing application-state workflows that reduced booking interaction steps by 30%.
- Optimized mobile data structures and asynchronous APIs, improving core application responsiveness by 19%.
- Collaborated with the Lead Mobile Architect to translate application designs into production-ready interfaces across 8+ core screens, maintaining consistent responsive behavior and component structure.
- Debugged mobile rendering, state-management, and API-integration issues across the booking application, reducing recurring development defects by 24% through systematic troubleshooting.

PROJECTS

Grounded-SAM2 LVIS Benchmark | PyTorch, Grounding DINO, SAM2, LVIS, CUDA

- Engineered a Docker-containerized open-vocabulary object-segmentation benchmarking pipeline using PyTorch, Grounding DINO, and SAM2 on a balanced 16,000-image LVIS subset spanning 1,200+ categories, automating GPU-based detection, mask generation, RLE encoding, and official metric evaluation; achieved 18.6% rare-category mAP in zero-shot evaluation versus 0.0% for a fine-tuned Mask R-CNN baseline.

Vision Transformer Activity Recognition | Python, TensorFlow, TensorFlow Hub, NumPy, Pandas, UCF-101

- Built a modular video-classification pipeline with NumPy/Pandas preprocessing and TensorFlow Hub Vision Transformers (ViT), comparing attention-based architectures with CNN baselines through configurable augmentation, hyperparameter tuning, and model-evaluation workflows. Co-authored and published peer-reviewed research in the Springer Book Series validating deep learning systems across diverse athletic **domains: [springer.com](https://www.springer.com)**

EDUCATION

Master of Applied Computer Science | Concordia University, Montreal, Quebec Sep 2024 – June 2026

Bachelor of Technology in Computer Engineering | CHARUSAT University, Gujarat, India July 2020 – June 2024